



Connectomic analysis of the central complex in fruit fly larvae



Laura Lungu

Supervisor: Professor Albert Cardona

Advisor: Dr. Marco Tripodi

Department of Physiology, Development and Neuroscience
University of Cambridge

This dissertation is submitted for the degree of
Doctor of Philosophy

October 2025

I would like to dedicate this thesis to my loving parents . . .

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 60,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Laura Lungu
October 2025

Acknowledgements

I would like to acknowledge Michael Clayton - help in data analysis and coding Nicolò Ceffa - help with light sheet imaging, flywork; Pedro Gomez Daniel Lalanti Venkat for teaching me foundations of flywork and immunostaining; Yijie Yin - help with neurotransmitter volumes Sam Harris - for fly genetics and behavioural testing

Abstract

Abstract

In holometabolous insects such as the fruit fly *Drosophila melanogaster*, the brain central complex (CX) develops during metamorphosis and serves the adult stage. Whether a form of the CX exists in the brain of the evolutionarily novel larval stages is not known. Here, we analyzed the connectome of the larval brain and, on the basis of neuronal lineages, synaptic connectivity patterns, and anatomy, identified a putative larval CX, comprising 4 key neuropils: the protocerebral bridge (PB), the ellipsoid body (EB), the fan-shaped body (FB) and the noduli (NO). Consistent with our interpretation, we found in the larval brain synaptic connectivity patterns characteristic of the adult, including (i) visual input into the PB and EB; (ii) modulation of CX neuropil inputs by the mushroom body (MB); (iii) reciprocal connectivity between CX neuropils and select MB compartments; and (iv) strong connectivity between CX neuropils. While some neuronal lineages contributing to the larval CX do not contribute to the adult CX, many others are conserved. The characterization of a larval CX brings structure to largely unexamined larval brain circuits, linking with a vast body of literature, and will inform the design of experiments to probe larval brain function.

Table of contents

List of figures	xiii
List of tables	xv
1 Discussion	1
1.1 Is there a Navigational Centre (Central Complex) in the Larval <i>Drosophila</i> ?	1
1.1.1 Larval CX presents important similarities to the Adult CX	2
PB	2
EB	2
FB	3
NO	4
1.1.2 Navigation, place learning, and memory	4
1.1.3 DANs in the FB and PB	5
1.2 Functions of the Larval Central Complex	6
1.2.1 Network connectivity predictions	6
1.2.2 The larval CX is responsive to light	7
1.2.3 A Multisensory Integration Centre in the Larva	7
1.2.4 Conclusion	9
References	11

List of figures

List of tables

Chapter 1

Discussion

1.1 Is there a Navigational Centre (Central Complex) in the Larval *Drosophila*?

Our inquiry into the nature of a fraction of neurons in the remaining *terra incognita* of the larval brain of the fruit fly yielded a series of neurons, neuropils and circuits with undeniable developmental, connectivity, and anatomical similarities to the set of core neuropils of the central complex of the adult fly brain. The key anatomical difference of split brain hemispheres and the very early arrest of neuroblast proliferation in the larvae of Diptera impose strong constraints as to the extent of the anatomical and cellular similarities between the larval and the adult brain.

Notwithstanding, most larval brain neurons survive metamorphosis, help articulate homologous brain neuropils in the adult (Prieto-Godino et al., 2012), and participate to neural circuits in the adult brain. For example, the command neurons for backward locomotion (Carreira-Rosario et al., 2018), and MB MBINs and MBONs (Truman et al., 2023).

Intriguingly, while we find that larval brain neuronal lineages contribute the same neuronal cell types to the same CX neuropils as their corresponding adult brain instances do (9 of 15 lineages), we also find a number of neurons contributing to the larval CX that originate in non-CX lineages. This pattern has been reported before for the MB (Truman et al., 2023), with some larval MB neurons later on either undergoing apoptosis or, astonishingly, contributing to CX neuropils after metamorphosis. With the MB being an undeniably homologous structure to that of the adult and the larval Kenyon cells making up the bulk of the adult fly brain's gamma Kenyon cells, the recruitment of non-CX lineages into contributing CX neurons in the larva is perhaps part of a common theme in an animal with a limited neuron budget. The broader context being that the larva of the holometabola is derived (Truman & Riddiford,

2019, 1999), an evolutionary novelty that enabled this group of animals to exploit novel ecological niches that, to boot, avoids competition with the adults. In such predicament, it is perhaps not surprising that other larval neuropils similarly to the MB temporarily adopt neurons as their own that will eventually either perish by apoptosis or return to an ancestral cell fate away from the CX.

1.1.1 Larval CX presents important similarities to the Adult CX

PB

The Protocerebral Bridge (PB) serves as a critical hub for processing navigational signals in both the larva and the adult *Drosophila*, with remarkable conservation of lineage origins and circuit architecture. In the adult, PEN (PB.d EB.bNO.b) columnar neurons are core components of the ring attractor network and they synapse bidirectionally onto PB horizontal neurons (EPG, EB.d PB.b GA.b) to adjust head direction based on angular velocity. The larval PB similarly features PB columnar neurons (a PB.d NO.b and a PB.d EB.d LAL.b) that collect axo-dendritic inputs from PB horizontal neurons (HF-PB) and are derived from homologous developmental lineages (DPMpm1/2, DPMm1, CM4).

Both developmental stages show tight integration between the PB and associative memory circuits: the adult PB receives modulatory dopaminergic (LPsP) and octopaminergic (P1-9) input, while larval PB horizontal fibres integrate multisensory information (visual and olfactory) pre-processed by MB-associated Convergence Neurons (CNs), in addition to receiving dopaminergic input via eDAN-1, eDAN-2, eDAN-6 and eDAN-7. This architectural conservation suggests the PB's fundamental role in coordinating navigational computations could have been maintained despite the dramatic reorganization of the nervous system during metamorphosis.

EB

The Ellipsoid Body (EB) in the larva receives neuronal contributions from several lineages that are homologous to adult CX lineages. Specifically, Ring neurons (RNs) originate from BAMv1/2 (adult LALv1), DALv23 (adult EBa1), and CM1/3 (adult DM5/6) lineages, whilst the DALv23 lineage additionally contributes wedge neurons.

The core connectivity between Ring and Wedge neurons within the larval EB is highly reminiscent of the adult circuit organization:

Wedge Neurons Network: Larval EB wedge neurons (W1–W8) form a densely recurrent network, resembling the adult wedge EPG neurons network. The predominant connection motif is axo-dendritic synapses, although weaker axo-axonic contacts are also present.

Wedge-to-Ring Connectivity: All eight larval wedge neuron pairs synapse directly onto several identified ring neuron (RN1/2a,b and RN3), creating an all-to-all Wedge to Ring connectivity motif that closely mirrors the characteristic pattern observed in the adult EB. However, these connections are exclusively axo-axonic, and may be indicative of a different functionality.

Ring Neurons Interconnectivity : Larval ring neurons (RN1/2, RN3) are reciprocally connected axo-axonically, a feature consistent with the dense interconnectivity seen among nearly all adult ring neuron types.

A further conserved feature from the adult is dopaminergic modulation of the wedge neurons. The adult EPG (wedge) neurons are modulated by dopamine (Frighetto et al., 2022), and anatomically we observe a direct dopaminergic synapse from eDAN-3 onto the larval W-1 wedge neuron. This conserved motif suggests neuromodulatory control of wedge circuitry across developmental stages. Targeted manipulations of eDAN-3 combined with calcium imaging of Wedge Neurons would test whether this input adjusts dynamics of the larval EB circuit.

FB

The larval FB shares several lineages with the adult FB neurons. The DM1-DM4 (larval lineages DPMm1, DPMpm1, DPMpm2, and CM4, respectively) lineages that contribute to the adult CX also generate a number of FB tangential neurons (hs-FBs) and 5 pairs of FB columnar neurons in the larva. In addition, the DALcm12 lineage (adult CREa1) contributes 1 pair of of columnar FB (FB.d5).

A key, evolutionarily conserved feature of the FB is its association with the Mushroom Body (MB), the center for learning and memory: in both the adult FB (via tangential neurons targeting middle layers) and the larval FB (via tangential neurons hs-FBs and intrinsic neurons), this neuropil has recurrent connections with MBONs. In the case of larva, majority of FB neurons receive indirect synaptic from MB related neurons (MB2ONs, MB2INs and CNs), nevertheless one direct connection exists from MBON-c1.

At both developmental stages, the FB is characterized by dense and highly recurrent networks composed of columnar cells (PB-FB-*, hDelta, vDelta types in the adult and FB.d in the larva) and tangential cells. However, the larval structure lacks the clear layering and high neuronal diversity seen in the adult. Intriguingly, adult Fan-Shaped Body (FB) tangential neurons (FBt) are known to synapse onto FB columnar neurons-notably vDelta, hDelta and PFL types-and this same tangential to columnar motif is evident in the larval FB. All identified hs-FB tangential neurons in the larva make synaptic contacts onto at least one FB.d columnar neuron type.

When it comes to neuromodulatory circuits, in the larva, we see FB tangential neurons, particularly hs-FB neurons, receive direct input from eDAN-2, eDAN-4l and eDAN-4m. Similarly in the adult 2 DANs (from PPL cluster) provide input to FB tangential neurons to regulate sleep-wake cycle (Liu et al., 2012).

Importantly, at both stages these neuropils function as essential hubs for multisensory integration, processing both visual inputs (via VPNs and VLNs in the larva, via POL neurons in the adult) as well as olfactory cues (often routed via the Lateral Horn and MB through Convergence Neurons) - see section ??.

NO

The most striking feature of the larval noduli is the morphology of its neurons' compact clutchy axons, which is a characteristic feature of the adult NO. Since the NO in the adult is by default split across the two hemispheres, it is actually plausible that such a structure could be anatomically conserved at an earlier stage of this animal (i.e. the larval brain commissure wouldn't be so disruptive for this particular neuropil). In addition, overlapping contributing developmental lineages include DM1 (DPMm1), DM2 (DPMpm1), DM3 (DPMpm2) and SIPp1 (DPMl12), supporting a shared ontogenetic origin with the adult NO, and the connectivity with LAL is present at both stages of this animal (however in the opposite direction with NO.d LAL.b in the larva and LAL.d NO.b in the adult).

1.1.2 Navigation, place learning, and memory

The central complex has been associated with place learning (for review see (Pfeiffer & Homberg, 2014)), with the EB in particular being necessary for visual place learning in flies (Ofstad et al., 2011). With fly larvae being capable of local search behaviour (Kromp et al., 2024), the structure of the larval central complex and its close interactions with the center for associative memory, the mushroom body, must be examined.

We reported that the 'c' compartment of the larval MB is intimately associated with the FB. Intriguingly, MBON-c1 lacks a known role in associative memory to date, with all tests having been performed for olfactory associative memory (Eschbach et al., 2020). Likewise, we discovered a tight loop between the NO and the MB vertical lobe, by means of neurons that also lack a known function in learning: the multi-compartment MBON-p1, whose dendrites span across the MB vertical lobe, and MBIN-11, a MB input neuron of unknown neurotransmitter signature and function. Both of these MB compartments ought to be examined experimentally for potential roles in navigation or place learning.

With the NO having a role in the adult fly brain controlling the time course of walking activity (Strauss & Heisenberg, 1993) and in handedness in locomotion (Buchanan et al., 2015), and the larval MB vertical lobe mediating aversive memories (Eschbach et al., 2020), our report of a tight loop between the output of the MB vertical lobe and modulatory input onto it via the NO may provide the necessary corollary discharge to adjust MB operations to imminent locomotor activity, potentially laterally biased since both MBON-p1 and MBIN-11 are entirely ipsilateral-only neurons.

Furthermore, we found that the 'g' compartment of the MB, whose MBON-g1 and MBON-g2 promote avoidance (Eschbach et al., 2021) and its DAN-g1 is implicated in aversive learning (Eschbach et al., 2020), is intimately and recurrently associated with the Wedge and Ring neurons of the EB. Two further MBONs, MBON-a2 and MBON-b3, likewise synapse onto EB neurons. Among these, the function of MBON-b3 remains unknown. Intriguingly, both MBON-a2 and MBON-b3 are rare among the MBONs in having an ipsilateral-only dendrite but a bilateral axon, when the arborization pattern of MBONs is most often the opposite: bilateral dendrites and an ipsilateral axon. The ability of these MBONs to differentially discern sensory inputs between the left and right sides of the body is suggestive of a role in lateralized behavioural responses, such as in taxis.

In loss-of-function experiments, the suppression of DAN-d1 with TNT removes excitatory drive from MBON-d1, which is GABAergic (Eschbach et al., 2021). From the connectome we interpret that a neuron postsynaptic to MBON-d1, MB2ON-63, which is a PB horizontal fiber distinct from DALv1, will then receive less GABAergic input. And therefore, MB2ON-63 receives unopposed excitatory input from visual PNs (PVL09) and other excitatory visual neurons (ChalOLP via 5thLN), and from MBON-c1, all of which are cholinergic (Eschbach et al., 2021; Larderet et al., 2017). We speculate that, if in wild-type conditions the inhibition from the MB via MBON-d1 was potentially subtracting an expected intensity of light, to perform a temporal comparison, without it the animal would no longer be able to assert where the light has increased or decreased, leading to the need for continuing to sample by head casting or turning.

1.1.3 DANs in the FB and PB

In the adult CX, dFB is a DAN that releases dopamine at the dorsal FB and is capable of triggering sleep (Pimentel et al., 2016). Intriguingly, we found a DAN (eDAN-2) among the FB intrinsic neurons, and three additional DANs (eDAN-4l, eDAN-4m) that synapse onto a number of FB.d and FB.b neurons. Identifying genetic driver lines for these neurons will enable testing experimentally their potential role in sleep regulation in larvae.

Notably, in the larva there is a dopaminergic neuron, eDAN-6, projecting its dendrites to the FB and axons to the PB (FB.d PB.b). This cell could be related to overall responsiveness and alertness of the animal, since there is a report that DANs projecting to the adult PB mediate increases in aggressiveness (Alekseyenko et al., 2013), and other DANs also projecting to the adult PB decrease sleep (Tomita et al., 2021). Similarly, the other eDANs sending synapses to the PB (eDAN-1, eDAN-2, eDAN-7) may be implicated in similar functions.

All of the identified non-MB DANs do await the identification of specific genetic driver lines and appropriate experimental setups to study their specific function, and their role in the larval central complex.

1.2 Functions of the Larval Central Complex

1.2.1 Network connectivity predictions

The current network connectivity suggests that visual input is highly concentrated in the central complex of the larva, and that this and the olfactory input are both processed before the decision to act on them. There seems to be a bias towards blue light processing, interestingly, since there are significantly more synapses from several more VPNs (90% blue sensing Rh5-PR input) than from VLNs (79% RH6-PR input). This implicates that the putative central complex might be involved in mediating escape behaviours.

Intriguingly, one of DN found downstream of the CX is MB2ON-39, also known as Ipsigoro, is known to mediate escaping behaviour (rolling) (Jones et al., 2025). MB2ON-39 was proved to integrate learnt information (via input from MBONs) as well as nociceptive information (via synapses from ascending neurons) to facilitate rolling via direct inputs to Goro (Burgos et al., 2018).

Another cell one layer away from the CX, MB2IN-172 is identified as a descending neuron (DN) in the subesophageal zone (DN-SEZ) originating from the DPMm2 lineage. This neuron depolarizes in response to nociceptive input and provides recurrent feedback to primary nociceptive interneurons (Basins), likely contributing to feedback modulation within the nociceptive network (Randel et al., 2025).

The coexistence of these two nociceptive pathways—one driving action (e.g., MB2ON-39) and the other shaping sensory gain (e.g., MB2IN-172)—implies that the CX contributes not only to the initiation of escape but also to its fine-tuning. This positions the larval CX as a higher-order hub that integrates aversive, learned, and sensory contexts to regulate behavioural output, paralleling the adult CX's role in adaptive navigation.

Two other descending neurons downstream of the CX, DMPI_ant_ven descending 3 and BAmD descending, project to the ventral nerve cord (DN-VNCs). Optogenetic activation screens have shown that DMPI_ant_ven descending 3 triggers backward locomotion, while BAmD descending initiates turning movements. While these responses suggest potential roles in escape behaviors, further investigation combining circuit analysis with whole-brain calcium imaging would be necessary to establish their specific functions in the context of CX-mediated behaviors.

1.2.2 The larval CX is responsive to light

In line with our intuition, calcium imaging data (Fig. ??) corroborate the electron-microscopy evidence that the larval central complex is responsive to blue light conveyed by RH5-PR. In vivo calcium recordings revealed a robust increased activity in PB neurons upon blue-light stimulation. These functional data mirror the anatomical convergence of RH5-pathway VPNs onto PB-projecting neurons and support the conclusion that the PB plays a prominent role in early processing of blue-light cues. However, further examination of other CX neuropils is needed for the complete picture, especially in the NO.b neurons, since 3 of them get direct input from VPNs.

In addition the DAN-d1 (EB.d), which collects visual input from 5thLN, is directly downstream of both blue and green sensing photoreceptors, and elicits a significant turning response in loss-of-function experiments when larvae are exposed to green light, further reaffirming the hypothesis that this brain region must be implicated in processing and regulating the animal's responses to visual stimuli.

1.2.3 A Multisensory Integration Centre in the Larva

Gepner et al. (Gepner et al., 2015) provided direct behavioural evidence that *Drosophila* larvae combine visual and olfactory gradients into a single, coherent sensory representation prior to committing to a navigational decision. In their key experiment larvae were presented with conflicting cues (aversive blue light versus optogenetic activation of appetitive ORNs) and showed no persistent contradiction between types of taxis, implying that multimodal information is resolved upstream of motor execution rather than by parallel, independent taxis programs.

This behavioural finding is a key indicator that there must be a region in the larval brain specifically responsible for integration and processing sensory stimuli into a coherent navigational outputs. I propose that the identified larval central complex constitutes a strong candidate for this role, with every neuropil identified (PB, EB, FB and NO) receiving visual

inputs from both visual projection neurons (VPNs, cells directly downstream of RH5 blue sensing photoreceptors) and visual local interneurons (VLNs, cells directly downstream of RH6 green sensing neurons; see section ??) as well as olfactory input via the MBONs, MB2ONs and CNs (see section ??), and modulatory signals from several DANs.

The PB in both the larval and adult brain exhibits multi-sensory integration capabilities. In the larva, horizontal PB fibres (DALv1 neurons) receive integrated sensory input via Convergence Neurons (CNs). These CNs (e.g., CN-53, CN-54) collect and combine information from both the Lateral Horn (innate responses) and MBONs (learned responses). Additionally, CN-54 and other CNs receive direct visual input from both VPNs and VLNs, establishing the PB as a hub for visual-olfactory integration. This multi-sensory convergence is further modulated by dopaminergic input from multiple eDAN neurons (eDAN-1, -2, -6, and -7). The EB incorporates multisensory information through multiple pathways. First, reciprocal connections exist between EB Ring Neurons (RNs) and the MB 'g' compartment, where MBON-g1 and -g2 modulate RN1/2. These RNs, along with RN3, receive indirect visual input from VPNs. Additionally, all four EB-projecting MBONs (MBON-g1, g2, a, b3) receive weak synaptic input from a single VLN, further integrating visual information into the circuit. Of course, all these MBONs are well known to be involved in olfactory input processing.

The FB forms a loop with the MB 'c' compartment via an FB.b (MB2ON-125), a neuron that sends axonal boutons to the FB. In addition, MB2ON-125 as well as several tangential hs-FB neurons collect visual input from VPNs via CNs. The FB's role in multisensory integration is further evidenced by its dense recurrent network of columnar and tangential neurons that process both input from visual neurons (VPNs and VLNs) and olfactory information routed through multiple MB output pathways. This integration is modulated by dopaminergic input from eDAN-2, eDAN-4l and eDAN-4m, suggesting state-dependent processing of multisensory information.

Lastly, the NO additionally integrates multiple sensory modalities. MBON-p1, a NO.b neuron, forms a recurrent network between the NO and MB vertical lobe by sending axonal outputs into this neuropil and synapsing onto several LAL.b neurons. Additionally, the NO is unique among CX neuropils in that neurons contributing axons to it receive direct visual input from VPNs (specifically nc1 and nc2 neurons), potentially suggesting a privileged role in early visual processing for navigation. Further investigation into these NO.b neurons, especially behavioural assays, is needed to establish a potential role in the visual processing system.

1.2.4 Conclusion

In conclusion, our interpretation of some of the until now under-examined larval brain circuits as a numerically reduced version of the adult fly central complex is coherent with the evolution of the larval stage in the Holometabola and with the known cell types as well as overall synaptic connectivity of the corresponding neuropils in adult fly brain, bringing a whole field of study into a life stage of reduced dimensions and numbers of neurons and synapses.

We now have the full wiring diagram of about 300 neurons comprising a region we determined as the central complex of the *Drosophila* larva, with four identified neuropils (PB, EB, FB, NO; Fig. ??), and their corresponding neuronal lineages (Table ??) as well as their neural types (Table ??). With a fully mapped circuitry of this area, we have a comprehensive understanding of the tight recurrent networks between and within the larval CX neuropils, as well as with the accessory structures such as the LAL. Crucially, we observe its high recurrence with the MB (implying some level of olfactory processing) as well as its involvement in visual input processing (via VPNs and VLNs), making it plausible that this area performs the sensory integration proposed by (Gepner et al., 2015). With the identified tight circuit loops between understudied MB compartments and the larval CX, an opportunity opens to examine the neural circuit basis of spatial navigation and place learning in this experimentally tractable animal.

In addition, we now have an inkling into the function of these circuits. With eight newly identified DAN neurons (eDANs) in the larval brain outside of the MB, and two of them being CX neurons, we have an increased understanding of the dopaminergic system and its relationship to the CX, posing the possibility of its modulation especially via the FB; not only that, loss-of-function behavioural assays suggest the relevance of a previously identified DAN-d1 (which sends dendrites to the EB) for larvae's behaviour in response to light. Last but not least, calcium activity recordings strongly support the role of the CX in light processing, with the PB neurons showing pronounced responses to blue light stimuli.

For the future, it's integral to confirm the role of this identified region with complementary functional data around the identified network. Ideally, lines would be found for all the 300 neurons, and future experimental setups would allow for multisensory testing during calcium activity monitoring, potentially fully replicating the gepner et al study.

References

- Alekseyenko, O. V., Chan, Y.-B., Li, R., & Kravitz, E. A. (2013). Single dopaminergic neurons that modulate aggression in drosophila. *Proceedings of the National Academy of Sciences*, 110(15), 6151–6156.
- Buchanan, S. M., Kain, J. S., & De Bivort, B. L. (2015). Neuronal control of locomotor handedness in drosophila. *Proceedings of the National Academy of Sciences*, 112(21), 6700–6705.
- Burgos, A., Honjo, K., Ohyama, T., Qian, C. S., Shin, G. J.-e., Gohl, D. M., Silies, M., Tracey, W. D., Zlatic, M., Cardona, A., et al. (2018). Nociceptive interneurons control modular motor pathways to promote escape behavior in drosophila. *Elife*, 7, e26016.
- Carreira-Rosario, A., Zarin, A. A., Clark, M. Q., Manning, L., Fetter, R. D., Cardona, A., & Doe, C. Q. (2018). Mdn brain descending neurons coordinately activate backward and inhibit forward locomotion. *Elife*, 7, e38554.
- Eschbach, C., Fushiki, A., Winding, M., Afonso, B., Andrade, I. V., Cocanougher, B. T., Eichler, K., Gepner, R., Si, G., Valdes-Aleman, J., et al. (2021). Circuits for integrating learned and innate valences in the insect brain. *Elife*, 10, e62567.
- Eschbach, C., Fushiki, A., Winding, M., Schneider-Mizell, C. M., Shao, M., Arruda, R., Eichler, K., Valdes-Aleman, J., Ohyama, T., Thum, A. S., et al. (2020). Recurrent architecture for adaptive regulation of learning in the insect brain. *Nature neuroscience*, 23(4), 544–555.
- Frighetto, G., Zordan, M. A., Castiello, U., Megighian, A., & Martin, J.-R. (2022). Dopamine modulation of drosophila ellipsoid body neurons, a nod to the mammalian basal ganglia. *Frontiers in physiology*, 13, 849142.
- Gepner, R., Mihovilovic Skanata, M., Bernat, N. M., Kaplow, M., & Gershow, M. (2015). Computations underlying drosophila photo-taxis, odor-taxis, and multi-sensory integration. *Elife*, 4, e06229.
- Jones, B. M., Harris, S. N., Ceffa, N. G., Cardona, A., & Zlatic, M. (2025). Descending neurons integrate learnt information from mushroom body with context to promote escape behaviour. *bioRxiv*, 2025–09.
- Kromp, J., Triphan, T., & Thum, A. S. (2024). Finding a path: Local search behavior of drosophila larvae. *bioRxiv*, 2024–11.
- Larderet, I., Fritsch, P. M., Gendre, N., Neagu-Maier, G. L., Fetter, R. D., Schneider-Mizell, C. M., Truman, J. W., Zlatic, M., Cardona, A., & Sprecher, S. G. (2017). Organization of the drosophila larval visual circuit. *Elife*, 6, e28387.
- Liu, Q., Liu, S., Kodama, L., Driscoll, M. R., & Wu, M. N. (2012). Two dopaminergic neurons signal to the dorsal fan-shaped body to promote wakefulness in drosophila. *Current Biology*, 22(22), 2114–2123.
- Ofstad, T. A., Zuker, C. S., & Reiser, M. B. (2011). Visual place learning in drosophila melanogaster. *Nature*, 474(7350), 204–207.

- Pfeiffer, K., & Homberg, U. (2014). Organization and functional roles of the central complex in the insect brain. *Annual review of entomology*, 59(1), 165–184.
- Pimentel, D., Donlea, J. M., Talbot, C. B., Song, S. M., Thurston, A. J., & Miesenböck, G. (2016). Operation of a homeostatic sleep switch. *Nature*, 536(7616), 333–337.
- Prieto-Godino, L. L., Diegelmann, S., & Bate, M. (2012). Embryonic origin of olfactory circuitry in drosophila: Contact and activity-mediated interactions pattern connectivity in the antennal lobe.
- Randel, N., Wang, C., Clayton, M. S., Wang, K., Pang, S., Xu, S. C., Champion, A., Hess, H. F., Cardona, A., Keller, P. J., et al. (2025). Combining brainwide activity imaging and electron microscopy reveals novel nociceptive circuits. *bioRxiv*, 2025–09.
- Strauss, R., & Heisenberg, M. (1993). A higher control center of locomotor behavior in the drosophila brain. *Journal of Neuroscience*, 13(5), 1852–1861.
- Tomita, J., Ban, G., Kato, Y. S., & Kume, K. (2021). Protocerebral bridge neurons that regulate sleep in drosophila melanogaster. *Frontiers in Neuroscience*, 15, 647117.
- Truman, J. W., Price, J., Miyares, R. L., & Lee, T. (2023). Metamorphosis of memory circuits in drosophila reveals a strategy for evolving a larval brain. *Elife*, 12, e80594.
- Truman, J. W., & Riddiford, L. M. (2019). The evolution of insect metamorphosis: A developmental and endocrine view. *Philosophical Transactions of the Royal Society B*, 374(1783), 20190070.
- Truman, J. W., & Riddiford, L. M. (1999). The origins of insect metamorphosis. *Nature*, 401(6752), 447–452.